Jatin Srivastava

AI Engineer | Machine Learning Engineer
+91-9569452553 | Noida, Uttar Pradesh, India
leetcode.com/u/jatinsri25 | srivastavajatin9898@gmail.com | linkedin.com/in/jatinsri25 | github.com/jatinsri25 | jatinsri.vercel.app

Summary

AI/ML Engineer with production experience in machine learning pipeline engineering, computer vision, and large language model (LLM) systems. Engineered ML data analysis and preprocessing pipelines driving measurable model improvements on NVIDIA's autonomous-vehicle perception project. Architected multi-agent Retrieval-Augmented Generation (RAG) platforms with high retrieval accuracy across 10,000+ documents. Proficient in PyTorch, LangChain, LangGraph, and FastAPI; focused on scalable MLOps and AI product innovation.

Skills

Languages: Python, JavaScript, TypeScript, SQL, C++

ML & AI: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, YOLOv8, Hugging Face Transformers, Deep Learning, NLP, Computer Vision

GenAI & LLMs: LangChain, LangGraph, RAG, Prompt Engineering, OpenAI API, Google Gemini API, FAISS, Vector Databases, Embeddings, Agent Orchestration

Backend & Databases: FastAPI, Node.js, Express.js, PostgreSQL, MongoDB, Redis, Neo4j

Frontend & DevOps: React.js, Next.js, Tailwind CSS, AWS, Docker, Git, GitHub, MLflow, JWT, RBAC

Experience

Zensar Technologies **Oct 2025 – Jan 2026**
ML Data Engineer & Analyst (NVIDIA Autonomous Vehicle Project) *Remote*

- Designed Python-based data analysis pipelines to profile, validate, and quality-check 50,000+ image samples used in NVIDIA's autonomous-vehicle perception model training.
- Developed statistical validation scripts to audit bounding-box label consistency across 12+ training splits, reducing systematic labeling errors by over 30%.
- Built OpenCV and YOLOv8 preprocessing pipelines automating frame extraction, bounding-box generation, and dataset versioning, cutting manual processing time by 60%.
- Analyzed model training metrics with NVIDIA ML researchers to detect class imbalances across 15+ object categories and refine labeling taxonomy for LiDAR and camera-fusion perception tasks.
- Delivered a 4-point F1-score improvement on downstream AV perception models by diagnosing and resolving data inconsistencies across the training corpus.

CMP Techseract LLP **Jun 2024 – Aug 2024**
Full-Stack Developer Intern *Lucknow, India*

- Shipped three MERN-stack applications to production serving 1,000+ daily active users.
- Halved average API response time from 400ms to under 200ms via MongoDB indexing and connection pooling.
- Increased user engagement by 22% by integrating payment gateways and notification services into the React.js frontend.

Projects

RAG-IT – Production-Grade RAG System for Codebase Q&A | *Python, ChromaDB, BM25, BGE Embeddings, Cross-Encoder Reranking, Gemini API, FastAPI* GitHub

- Architected a hybrid retrieval pipeline fusing dense BGE embeddings with BM25 keyword search via Reciprocal Rank Fusion, fixing dense-only failures on exact code identifiers (function names, error codes, config keys).
- Implemented Corrective RAG (CRAG) with LLM-based relevance grading and automatic query rewriting, letting the system self-detect low-confidence retrievals and recover instead of hallucinating.
- Added cross-encoder reranking (ms-marco-MiniLM-L-6-v2) and HyDE query expansion, improving answer faithfulness and citation accuracy over vector-only retrieval baselines.
- Built dual-layer guardrails (PII redaction and prompt-injection detection) at ingestion and inference time, and validated output quality with a combined RAGAS and LLM-as-judge evaluation harness.

Prayaas – Civic Tech AI Platform | *Next.js, FastAPI, Google Gemini API, Redis, PostgreSQL* GitHub

- Built a 6-stage NLP pipeline converting citizen voice and text reports into structured JSON tickets.
- 6-stage pipeline—audio transcription, prompt-injection guard, PII redaction, token budgeting, LLM formatting, and solution generation—processes 500+ citizen reports per month with full end-to-end automation.
- Engineered a Gemini fallback chain with auto-retry on 429 and 503 errors, achieving 99%+ request success rate across API calls.

QuirkQuack – Real-Time Chat Application | *React.js, Node.js, Socket.io, MongoDB, Redis, JWT* GitHub

- Launched a WebSocket-driven chat platform handling 100+ concurrent connections with sub-50ms message delivery.
- Secured WebSocket handshakes with JWT validation and scaled across Node.js instances using a Redis Pub/Sub adapter.

Education

Ajay Kumar Garg Engineering College **Ghaziabad, India**
Bachelor of Technology in Information Technology *2021 – 2025*

Little Flower School (ISC Board) **Gorakhpur, India**
PCM with Computer Science *2020*